

MD Minhazur Rahman

Data Scientist · Machine Learning Engineer

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Summary

Data scientist and ML engineer with an MSc in Data Science (University of Greenwich) and 3+ years of industry data and IT experience. I ship production-grade machine learning end to end — forecasting, imbalanced classification, and LLM/RAG — with two models live on GCP Cloud Run, deployed by Terraform with CI/CD and monitoring. First-author DOI-indexed preprint. Open to Data Scientist / ML Engineer roles worldwide with relocation and visa sponsorship.

Selected Projects (live, deployed)

Real-Time Demand Forecasting — Production MLOps Pipeline

2026

LightGBM · FastAPI · Docker · Terraform · Cloud Run

github.com/minhazda/synthetic-retail-mlops-pipeline

- **Cut forecast error 40.8% MAE / 47.3% RMSE vs. a seasonal-naive baseline** by building a typed, tested LightGBM pipeline with a single shared feature module that eliminates training/serving skew.
- **Deployed a live, public inference API on GCP Cloud Run** provisioned entirely as code (Terraform) with keyless GitHub→GCP CI/CD (Workload Identity Federation — no stored secrets).
- **Added production observability** with Prometheus metrics, a Grafana dashboard, and Evidently data-drift reporting on the running service.

Real-Time Card-Fraud Detection — Imbalanced Classification

2026

LightGBM · scikit-learn · FastAPI · Docker · Terraform · Cloud Run

github.com/minhazda/fraud-detection-mlops

- **Reached ROC-AUC 0.90 and PR-AUC 0.49 (~8× the base rate) on held-out data** by handling class imbalance (`scale_pos_weight`) and tuning the decision threshold on a separate validation split.
- **Served real-time scoring with built-in explainability** — engineering features from raw transactions at request time and surfacing the risk signals behind each decision in an interactive demo UI.

Demand Forecasting on Real Retail Data (UCI Online Retail II)

2026

pandas · LightGBM · DuckDB · time-series

github.com/minhazda/real-retail-forecasting

- **Improved pooled MAE 26.3% over a seasonal-naive baseline (beating it on 47/49 product series)** by documenting defensible cleaning of 1.07M real transactions and reproducing the findings in DuckDB SQL.

Privacy-Preserving RAG Agent — LLM Tool-Calling

2026

LangGraph · ChromaDB · on-device embeddings · FastAPI

github.com/minhazda/privacy-preserving-rag-agent

- **Shipped a tool-calling RAG agent behind a fail-closed privacy guard** (on-device embeddings, PII redaction) and built a two-tier evaluation: a deterministic CI gate plus a Claude LLM-as-judge harness with Langfuse tracing.

Professional Experience

GPH Ispat Limited

Feb 2022 – Aug 2024

Officer — Information Technology Department

Chittagong, Bangladesh

- **Automated departmental reporting and operational workflows** by developing internal Python and SQL tools, reducing manual effort across materials-management and reporting processes.
- **Sustained daily operational continuity for enterprise systems** used across departments at one of Bangladesh's largest publicly listed (DSE) steel manufacturers, through proactive maintenance and troubleshooting.
- **Safeguarded data integrity across inventory tracking** by monitoring and reviewing materials-management data feeding company reporting.

Vcube Soft & Tech

2021 – Feb 2022

QA Engineer — Computer Vision Data

Chittagong, Bangladesh

- **Raised training-data quality for production 3D floor-plan reconstruction models** by curating and validating annotated 360° image datasets and enforcing metadata consistency.
- **Built QA validation pipelines** that systematically caught annotation errors before they reached model training.

Education

MSc Data Science — Pass with Merit

Sep 2024 – Jan 2026

University of Greenwich

London, United Kingdom

- Dissertation (70%): real-time retail demand forecasting on privacy-preserving synthetic data — 37% MAE reduction; published as a first-author DOI-indexed preprint (Zenodo).
- Modules: Applied Machine Learning 71%, Cloud Computing & HPC 74%, Big Data Analytics, Statistical Methods for Time Series.

BSc Computer Science & Engineering

Independent University, Chittagong

Jan 2017 – Jan 2021

Chittagong, Bangladesh

- Final-year project: rare-event detection in surveillance video using 3D-ResNet18 and a custom CNN (PyTorch) on the UCF Crime dataset.

Technical Skills

Languages	Python, SQL, C++, R
ML / DS	scikit-learn, LightGBM, XGBoost, Random Forest, PyTorch; time-series forecasting, imbalanced classification, feature engineering
MLOps / Cloud	Docker, Terraform (IaC), GCP Cloud Run, GitHub Actions (CI/CD), MLflow, Prometheus, Grafana, Evidently
LLM / GenAI	LangGraph, RAG, ChromaDB, LLM-as-judge evaluation, Langfuse
Serving / Data	FastAPI, pandas, NumPy, DuckDB; rolling-origin cross-validation, model evaluation

Publications, Awards & Research

- **Publication:** Rahman, M. M. (2026). “Synthetic Retail Time-Series for Privacy-Preserving Demand Forecasting.” Preprint, Zenodo. DOI: [10.5281/zenodo.19479285](https://doi.org/10.5281/zenodo.19479285).
- **HPC research:** parallel 2D Jacobi solver in C++/OpenMP on an Azure HPC cluster — $4.81\times$ speedup on 8 threads.
- **Awards:** Winner, Game of Presentations 2019 (inter-university); Second Runner-Up, City Bank Bankers’ Hunt 2020 (national case competition).
- **Teaching:** supplementary ML instruction to 10+ MSc peers; 4 years of independent STEM/Python tutoring (15+ students).

Reference

Dr. Tuan Vuong — MSc Dissertation Supervisor, University of Greenwich · T.Vuong@greenwich.ac.uk. Further references available on request.